

# Housing Access and Fertility

## Compact Theory and Preliminary Quantitative Results

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### Abstract

Children require goods and housing space. Family-sized rentals are limited, while access to owner housing requires liquid wealth and borrowing capacity. Older households meanwhile hold much of the large-home stock. I study how this allocation of space changes the private price of children. A compact overlapping-generations model expresses rental limits, down-payment constraints, and old-owner retention as goods-equivalent wedges. It separates changes in family size among incumbent households from changes due to housing-market entry and tenure switching. A finite-life quantitative model adds income risk, saving, tenure and house-size choice, transaction costs, bequests, and children aging out of the household. The current calibration suggests that access to family-sized space is valuable to households near the fertility margin, while the timing of purchases is central to the response of births to housing policy.

### Status of this draft

This document is an extended research update rather than a completed paper. The compact model has been checked independently, and the results stated below include the assumptions needed for the derivations. The quantitative sections report a July 9, 2026 calibration snapshot. Two implementation corrections are underway: current housing choices must enter several moments contemporaneously, and the first age cell must use the intended earnings profile. Both corrections affect targeted moments, so the calibration and all counterfactual magnitudes remain provisional until the model is re-estimated.

The quantitative environment below follows the specification used to produce the snapshot: the owner premium applies to usable housing space, bequest utility is not normalized across family-size states, and all owners face the same debt floor. The policy exercises are preliminary patterns from an internally consistent diagnostic runtime. They will be rerun from the corrected and re-estimated baseline.

## 1 Introduction

A child changes a household's housing problem. It is not only another set of goods to buy, or more time to spend at home. It is often a need for a different dwelling: another bedroom, a larger living space, or a unit where raising a child is feasible. In many high-cost cities this margin is hard to adjust. Large rental units are scarce, family-sized homes are concentrated in ownership, and ownership requires down payments and borrowing capacity at ages when many households have little wealth.

This paper studies whether those space and financial constraints affect final family size and the number of births generated by a cohort. That distinction matters. A housing shock can move births across years without changing completed fertility. It can also change local births by changing who enters a high-cost housing market, even if completed fertility among incumbent households moves little. The question here is when housing constraints change completed fertility, when they mainly change entry and local scale, and how both effects depend on who can access family-sized space over the lifecycle.

The mechanism has three parts. First, children raise the household’s demand for space. If the rental market does not offer sufficiently large units, the shadow price of an additional room can exceed the posted rent. Second, the natural substitute is ownership, but buying a family-sized home requires liquidity and credit. A young household may value the larger unit and still be unable to purchase it. Third, older owners may remain in large homes even when their own marginal value of space is low, because transaction costs and tax rules subsidize staying put. Housing scarcity then shows up not only as a high price of space, but also as a mismatch between who occupies family-sized units and who values them for fertility.

A compact two-period model expresses the rental space constraint, the ownership finance constraint, old-owner lock-in, and the fertility distortion as goods-equivalent wedges. I then embed the same economics in a finite-life quantitative model. The model combines lifecycle fertility choice (Sommer, 2016; Doepke and Kindermann, 2019) with housing tenure choice, collateral constraints, and owner-occupied housing (Henderson and Ioannides, 1983; Sommer et al., 2013; Sommer and Sullivan, 2018). The compact housing block follows the one-market floorspace logic in Coven et al. (2025): all housing services clear in one market, while tenure, borrowing, and tax rules determine who can actually occupy family-sized units.

The current quantitative exercise is informative about the mechanism but not yet a final estimate of its size. The July snapshot places substantial fertility-response weight on households constrained by the rental-size limit, and it produces a sharp coincidence of births and purchases near the end of the fertile window. It also misses early ownership, young liquid wealth, and old-age exit from ownership. The first two misses are affected by the implementation corrections described above. I therefore use the current calibration to organize the evidence and the next estimation step, not to make a settled quantitative claim.

## 2 Related Literature

The paper connects work on fertility choice to work on housing tenure and the allocation of large homes. Since Becker (1960) and Becker and Lewis (1973), economics has treated children as choice variables that enter preferences and require resources. Quantitative versions place that choice in lifecycle environments with income risk, saving, timing, and intrahousehold bargaining (Jones et al., 2010; Sommer, 2016; Doepke and Kindermann, 2019). Recent work emphasizes that the U.S. fertility decline is difficult to explain with one postponement margin or period shock (Kearney et al., 2022).

Housing prices, housing wealth, and mortgage finance also move fertility behavior. Dettling and Kearney (2014) find opposite house-price effects for owners and nonowners, Lovenheim and Mumford (2013) find that housing wealth raises homeowners’ fertility, and Cumming and Dettling (2024) show that mortgage-rate pass-through affects births through cash flow. Hacamo (2021) and Dettling and Kearney (2025) connect mortgage access to home purchases and childbearing. In equilibrium, Couillard (2025) show that the supply of large units matters for aggregate fertility.

The outcome here is closer to completed fertility than to a current birth rate. [Simon and Tamura \(2009\)](#) find that rents per room predict lower fertility, including in completed-fertility specifications. [Clark \(2012\)](#) provides an important caution: expensive housing markets delay first births but do not clearly reduce completed fertility. More recent evidence links housing access to both timing and final family size ([Japaridze and Sayour, 2024](#); [Dettling and Kearney, 2025](#); [van Doornik et al., 2025](#)). I use this evidence to keep three objects separate: current births, completed fertility among incumbent households, and cohort births after housing-market entry responds.<sup>1</sup>

The closest housing-allocation paper is [Coven et al. \(2025\)](#). Their model studies how property taxes, capitalization, and financial constraints allocate housing between older homeowners and younger constrained households. I add children to that allocation problem. Family-sized space held by an incumbent can then change the private price of children for a young household. The retention force also relates to mortgage lock-in and housing ladders ([Fonseca and Liu, 2023](#); [Fonseca et al., 2026](#)). My baseline quantitative model uses transaction costs rather than fixed-rate mortgage lock-in, but the allocation logic is similar.

The quantitative housing block builds on models in which tenure, collateral constraints, and owner-occupied housing shape household behavior ([Henderson and Ioannides, 1983](#); [Sommer et al., 2013](#); [Sommer and Sullivan, 2018](#)). It also relates to spatial work in which housing supply and local prices allocate households across markets ([Rosen, 1979](#); [Roback, 1982](#); [Gyourko et al., 2013](#); [Hsieh and Moretti, 2019](#)). The present model keeps one housing market so that tenure, family size, and the allocation of space remain the central margins.

## Part I

# Compact Analytical Model

## 3 Environment

**Agents.** The economy is populated by overlapping generations of young and old households. A young household has type

$$i = (y_i, a_i),$$

where  $y_i > 0$  is disposable income and  $a_i \geq 0$  is entry wealth, inherited from the previous old generation. Conditional on entry, the young household chooses tenure, housing, completed fertility, and saving. Old households are indexed by  $j$ . An old incumbent is last period's young owner and enters old age with financial wealth, the house she bought, and an accrued taxable gain. She chooses consumption, how much housing to retain, and the estate she leaves at exit. An old renter carries no housing and no capital-gains wedge.

**Masses and demography.** At the start of period  $t$ , a mass  $M_t$  of potential young households is drawn from a distribution  $G_{Y,t}$ . A potential young household either enters the modeled housing market or takes an outside option. Nonentrants live outside the modeled market in both periods. Current young entrants become next period's old households, and the next potential-young cohort is formed by births plus a renewal inflow  $R_t$ . The stationary cohort mass is denoted by  $\bar{M}$ .

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<sup>1</sup>Life-course evidence documents that fertility, homeownership, dwelling size, and residential transitions move together ([Mulder and Billari, 2010](#); [Kulu and Steele, 2013](#); [Chudnovskaya, 2019](#)). The model treats those joint transitions as economic choices over the lifecycle.

**Preferences.** Children raise both consumption needs and housing needs. In the compact model fertility is continuous. A young household choosing consumption  $c_i$ , housing  $h_i$ , and fertility  $n_i$  has utility

$$u_i^Y(c_i, h_i, n_i) = \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i, \quad \alpha, \vartheta, \chi, \kappa > 0. \quad (1)$$

The term  $\chi n_i$  is the goods requirement of children and  $\kappa n_i$  is their housing-space requirement. A marginal child therefore uses the same housing stock as the parents.

Old households value consumption, housing, and bequests:

$$u_i^O(c_j^O, h_j^O, e_j) = \log c_j^O + \gamma \log h_j^O + \mathcal{B}(e_j), \quad \gamma > 0, \quad (2)$$

where  $e_j \geq 0$  is the estate and  $\mathcal{B}$  is increasing and concave, possibly identically zero. Bequest motives and housing utility in old age are preferences, not wedges. The closed-form fertility comparison below either uses  $\mathcal{B}(e) = b \log e$  with an interior bequest or sets  $\mathcal{B} \equiv 0$ ; the allocation wedge itself does not require that specialization.

**Housing.** There is a fixed stock  $\bar{H}$  of divisible floorspace. Households occupy floorspace by renting or owning. The two tenure modes have different size limits:

$$\mathcal{H}^R = \{h : 0 < h \leq \bar{h}^R\}, \quad \mathcal{H}^O = \{h : 0 < h \leq \bar{h}^O\}, \quad 0 < \bar{h}^R < \bar{h}^O. \quad (3)$$

This is the family-housing segmentation: small units are available to rent, but large family-capable units are mainly reached through ownership. All floorspace trades in one market. The user cost of housing is  $q$  and the asset price is  $P$ , linked by

$$q = \rho P, \quad \rho = r + \delta + \tau^P, \quad (4)$$

where  $r$  is the interest rate,  $\delta$  is depreciation, and  $\tau^P$  is the property-tax rate. For a given user cost  $q$ , a higher property tax lowers the asset price  $P$  and therefore lowers the down payment needed to buy a given house.

**Financing.** A buyer can finance at most a share  $\phi \in (0, 1)$  of the purchase price. Buying  $h$  units therefore requires the cash down payment

$$(1 - \phi)Ph. \quad (5)$$

Renters cannot borrow. Households can save in a bond at return  $1 + r$ . The compact model treats the down-payment constraint as an access constraint: it limits the amount of owner housing a young household can occupy.

**Capital-gains taxation and step-up basis.** Selling a house triggers a capital-gains tax. Dying with the house gives a stepped-up basis, so the accrued gain is forgiven. In a stationary real model, the gain comes from nominal price growth. If dollar prices grow at rate  $\varpi$ , a house bought one period ago has a taxable paper gain even when its real value is unchanged. The real flow-equivalent tax wedge per unit sold is

$$\bar{\ell} = \tau^g \frac{P\varpi}{1 + \varpi}, \quad (6)$$

after normalizing the period conversion factor to one. The wedge is a private tax liability avoided by holding the house until death. It is neither a preference nor a resource cost.

## 4 Households

Conditional on entry, young household  $i$  chooses a tenure mode  $m \in \{R, O\}$ , housing, fertility, consumption, and savings. For  $m \in \{R, O\}$ ,

$$W_i^m(q) = \max_{c_i, h_i, n_i, a'_i} \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i + \beta [\mathbf{1}\{m = R\}V^R(a_{j'}) + \mathbf{1}\{m = O\}V^O(a_{j'}, h_i)] \quad (7)$$

subject to

$$c_i + qh_i + a'_i = y_i + a_i, \quad (8)$$

$$h_i \in \mathcal{H}^m, \quad a'_i \geq 0,$$

$$\mathbf{1}\{m = O\}(1 - \phi)Ph_i \leq a_i, \quad (9)$$

$$c_i - \chi n_i > 0, \quad h_i - \kappa n_i > 0.$$

The young household's choices generate the old state. A renter carries bond wealth only. An owner carries both the house and the mortgage position:

$$a_{j'} = (1 + r)a'_i - \mathbf{1}\{m = O\}qh_i, \quad H_{j'}^0 = \mathbf{1}\{m = O\}h_i. \quad (10)$$

This transition makes the future tax liability part of the current owner choice rather than an exogenous tenure shifter.

Using  $P = q/\rho$ , the largest owner house feasible for type  $i$  is

$$H_i^O(q) = \min \left\{ \bar{h}^O, \frac{a_i \rho}{(1 - \phi)q} \right\}, \quad H_i^R = \bar{h}^R. \quad (11)$$

$H_i^m$  is the effective family-housing cap. A renter is capped by the rental stock. A buyer is capped by the owner size limit or by collateral, whichever is tighter.

An old incumbent enters with financial wealth  $a_j$ , carried housing  $H_j^0$ , and an accrued gain. She chooses consumption, retained housing, and an estate:

$$V^O(a_j, H_j^0) = \max_{c_j^O, h_j^O, e_j} u^O(c_j^O, h_j^O, e_j) \quad (12)$$

$$\text{s.t.} \quad c_j^O + e_j + qh_j^O = a_j + qH_j^0 - \bar{\ell}(H_j^0 - h_j^O), \quad 0 \leq h_j^O \leq H_j^0.$$

Selling one unit nets  $q - \bar{\ell}$ , so stepped-up basis acts as a subsidy to retaining the house. An old renter has no carried housing and no tax:

$$V^R(a_j) = \max_{c_j^O, h_j^O, e_j} u^O(c_j^O, h_j^O, e_j) \quad \text{s.t.} \quad c_j^O + e_j + qh_j^O = a_j, \quad 0 < h_j^O \leq \bar{h}^R. \quad (13)$$

**Assumption 1** (Regularity at the saving and retention margins). *The gains wedge satisfies  $0 \leq \bar{\ell} < q$ . A generated old owner has positive liquidation wealth, saves in youth, and chooses retention in the interior:  $a'_i > 0$ ,  $a_{j'} + (q - \bar{\ell})H_{j'}^0 > 0$ , and  $0 < h_{j'}^O < H_{j'}^0$ . Whenever an old-renter first-order condition is used, that housing choice is also interior.*

The assumption keeps the marginal owner price below smooth. At a saving or retention corner, the wedge decomposition uses a type-specific continuation price rather than the closed form in eq. (20).

## 5 Entry and Equilibrium

A potential young household can enter the housing market or take an outside option with value  $\bar{W}^E$ . Entry has Type-I extreme-value shocks with scale  $\kappa_E$ . The inside value is

$$W_i^Y(q) = \max\{W_i^R(q), W_i^O(q)\},$$

and the entry probability is

$$\pi_i^E(q) = \frac{\exp\{W_i^Y(q)/\kappa_E\}}{\exp\{W_i^Y(q)/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}}. \quad (14)$$

Let  $n_i^Y(q)$  be fertility in the inside tenure chosen by type  $i$ , and let  $\bar{n}_i^E$  be fertility in the outside option. Births generated by the potential-young cohort are

$$N(q) = \bar{M} \int [\pi_i^E(q)n_i^Y(q) + (1 - \pi_i^E(q))\bar{n}_i^E] dG_Y(i). \quad (15)$$

The baseline normalization sets  $\bar{n}_i^E = 0$ , but the entry-margin terms below are written so that outside fertility can be restored by replacing inside fertility with the gap  $n_i^Y - \bar{n}_i^E$ . In a stationary demographic equilibrium, births and renewal close the young cohort:

$$N + \bar{R} = \bar{M}. \quad (16)$$

The local formulas below hold  $(\bar{M}, G_Y, F_O, F_R, \bar{R})$  fixed. A full demographic counterfactual instead recomputes the stationary fixed point, or else states which object is held fixed and which object absorbs the birth change.

Let  $F_O$  and  $F_R$  be the stationary measures of old incumbents and old renters. Aggregate floorspace demand is

$$H^D(q) = \bar{M} \int \pi_i^E(q)h_i(q) dG_Y(i) + \int h_j^O(q) dF_O(j) + \int h_j^O(q) dF_R(j). \quad (17)$$

The compact equilibrium clears the single stock:

$$H^D(q) = \bar{H}, \quad P = q/\rho. \quad (18)$$

Stationarity requires current young owners to become next period's old incumbents, current young renters to become old renters, and births plus the renewal inflow to reproduce the next young cohort. The transition rules map old estates into entry wealth. They otherwise remain general, and the local results condition on the stationary cross-section.

## 6 Equilibrium Wedges

The model's sufficient statistics are marginal values of space in consumption goods. Let  $\lambda_i^m$  be the multiplier on the young household's budget,  $\theta_i^m$  the multiplier on the size limit, and  $\mu_i$  the multiplier on the owner's down-payment constraint. For renters,

$$\frac{\alpha(c_i^R - \chi n_i^R)}{h_i^R - \kappa n_i^R} = q + \zeta_i^R, \quad \zeta_i^R = \frac{\theta_i^R}{\lambda_i^R} \geq 0. \quad (19)$$

A capped renter values one more unit of housing above the rent it would command.

For young owners, anticipation of the future capital-gains liability raises the effective price of owning family-sized space. Under Assumption 1,

$$q^O = q + \frac{\bar{\ell}}{1+r}. \quad (20)$$

The owner access gap is

$$\zeta_i^O = \underbrace{\frac{\mu_i}{\lambda_i^O}(1-\phi)P}_{\zeta_i^{O,F}, \text{ down-payment component}} + \underbrace{\frac{\theta_i^O}{\lambda_i^O}}_{\text{owner-size component}} \geq 0, \quad (21)$$

so the owner's marginal value of space is

$$\frac{\alpha(c_i^O - \chi n_i^O)}{h_i^O - \kappa n_i^O} = q^O + \zeta_i^O. \quad (22)$$

When the owner size limit is slack,  $\zeta_i^O = \zeta_i^{O,F}$  and the gap is a financing gap.

The old incumbent's interior retention condition is

$$\frac{\gamma c_j^O}{h_j^O} = q - \bar{\ell}. \quad (23)$$

She keeps marginal space she values below the market user cost because selling it realizes the taxable gain. An old renter has no such wedge and, at an interior choice, values space at  $q$ .

The fertility condition prices the marginal child in goods. In tenure mode  $m$ , with  $q^R = q$  and  $q^O$  given by (20),

$$\underbrace{\frac{\vartheta(c_i^m - \chi n_i^m)}{n_i^m}}_{\text{willingness to pay for a marginal child}} = \underbrace{\chi + \kappa(q^m + \zeta_i^m)}_{\text{full price of a child}}. \quad (24)$$

A child's space requirement is priced at the household's own shadow value of space. A binding access constraint therefore acts as an implicit child tax  $\kappa \zeta_i^m$ .

## 7 Efficiency and Fertility

The planner comparison holds the stationary cross-section, entry probabilities, tenure assignments, savings, and real size-limit constraints fixed at the instant of the variation. The planner counts true household preferences and true resource costs. It does not count the avoided capital-gains tax as a social benefit.

**Proposition 1** (Local young-old housing misallocation). *Consider a stationary equilibrium. Hold entry, tenure assignments, savings, and real size-limit constraints fixed. Take a young owner whose owner size limit is slack and whose financing constraint binds, so  $\zeta_i^{O,F} > 0$ , and an old incumbent at an interior retention choice. If one unit of existing housing can be reassigned from the old incumbent to the young owner at zero real implementation cost, the planner's goods-equivalent surplus is*

$$\underbrace{(q^O + \zeta_i^{O,F})}_{\text{young buyer's marginal value}} - \underbrace{(q - \bar{\ell})}_{\text{old incumbent's marginal value}} = \zeta_i^{O,F} + \frac{\bar{\ell}}{1+r} + \bar{\ell}. \quad (25)$$

If this expression is positive, the competitive allocation does not solve the planner's conditional allocation problem.

The market price cancels. What remains are the young household's financing gap, the buyer's anticipated gains tax, and the old incumbent's realized gains tax. The two tax terms are transfers, not resource costs when fiscal revenue is returned lump sum. If implementing the reassignment requires real cost  $r_i^F$  per unit, that cost subtracts from the surplus. If the down-payment constraint is a primitive technological enforcement constraint and no instrument can relax it, the variation is not feasible.

Old occupancy is not inefficient by itself. An old owner who keeps a large house because she values it, or because she values leaving bequests, is using housing for reasons the planner also counts. The inefficiency is the part of retention supported by the forgiveness of capital-gains taxes at death, paired with young households who value family-sized space above its market user cost.

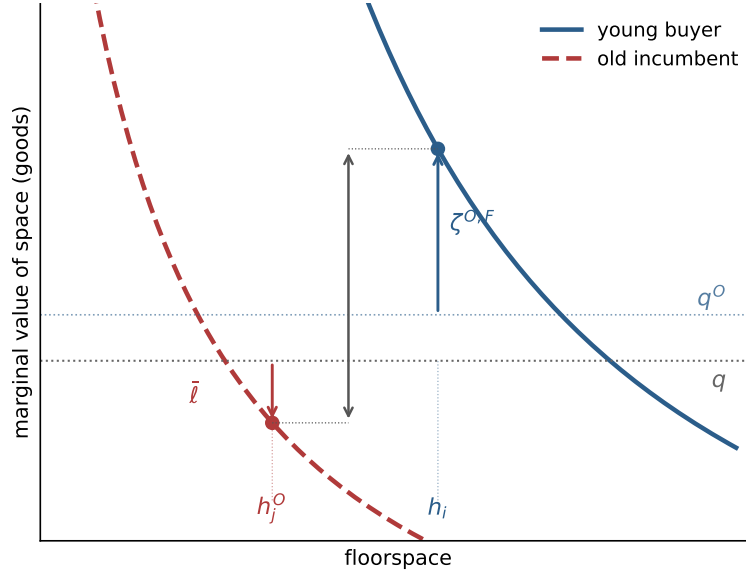


Figure 1: Local misallocation. A collateral-capped young buyer values marginal space at  $q^O + \zeta^{O,F}$ , while a locked-in old incumbent values retained space at  $q - \bar{\ell}$ . The vertical distance is the goods-equivalent surplus from a compensated marginal reassignment.

The same cap also determines fertility. Fix a tenure mode and a household whose housing is pinned at the effective family-housing cap  $H$ . For this comparison, suppose saving can be summarized by a constant load  $k \geq 0$  on residual adult consumption:

$$(1+k)(c - \chi n) + \chi n + q^m H = w, \quad w = y + a.$$

The log-bequest case  $\mathcal{B}(e) = b \log e$  gives  $k = \beta(1 + \gamma + b)$ ; with no bequest motive, set  $b = 0$ . After substituting this saving rule into the budget, fertility along the binding cap solves

$$\frac{\vartheta}{n} = \frac{(1+k)\chi}{w - q^m H - \chi n} + \frac{\alpha\kappa}{H - \kappa n},$$



Differentiating with respect to the cap gives

$$\frac{dn}{dH} = \frac{\frac{\alpha\kappa}{(H - \kappa n)^2} - \frac{(1+k)\chi q^m}{(w - q^m H - \chi n)^2}}{\frac{\vartheta}{n^2} + \frac{(1+k)\chi^2}{(w - q^m H - \chi n)^2} + \frac{\alpha\kappa^2}{(H - \kappa n)^2}}. \quad (26)$$

Relaxing access raises fertility when the space-relief force dominates the budget-tightening force. A useful sufficient condition along every binding cap is

$$\chi \leq \frac{(1+k)\kappa q^m}{\alpha}. \quad (27)$$

The derivative connects the misallocation wedge to fertility: policies that raise the effective family-housing cap matter only for households whose binding constraint they actually relax.

This household derivative aggregates into a local sufficient statistic for births. Let  $z$  be a policy parameter. For renters,  $H_i^R = \bar{h}^R$ ; for owners,

$$H_i^O(q) = \min \left\{ \bar{h}^O, \frac{a_i \rho}{(1-\phi)q} \right\}.$$

Let  $\mathcal{E}_m(z)$  be the set of types for whom tenure mode  $m$  is the unique active tenure, the effective cap  $H_i^m$  is strictly binding, and the binding component of the cap is locally unique. The pass-through of policy  $z$  to the effective family-housing cap is

$$\mathcal{P}_i^m(z) = \frac{\partial H_i^m(z)}{\partial z}.$$

For an exposed type, write residual adult consumption and residual space as

$$x_i^m = c_i^m - \chi n_i^m, \quad s_i^m = h_i^m - \kappa n_i^m.$$

The fixed-mode fertility response to one more unit of permitted family housing is

$$\Psi_i^m = \frac{\frac{\alpha\kappa}{(s_i^m)^2} - \frac{\chi q^m}{(1+k)(x_i^m)^2}}{\frac{\vartheta}{(n_i^m)^2} + \frac{\chi^2}{(1+k)(x_i^m)^2} + \frac{\alpha\kappa^2}{(s_i^m)^2}}. \quad (28)$$

The value of relaxing the cap in utility units is

$$\Theta_i^m \equiv \frac{\partial W_i^m}{\partial H_i^m} = \frac{\alpha}{s_i^m} - \frac{q^m}{x_i^m} = \frac{\zeta_i^m}{x_i^m}. \quad (29)$$

Thus the same access gap that measures misallocation also governs the entry response.

**Proposition 2** (Local fixed-state fertility statistic). *Holding prices, tenure choices, active sets, outside fertility, and the stationary demographic state fixed, the local effect of a cap-shifting policy on cohort births is*

$$\left. \frac{dN}{dz} \right|_{q, \text{tenure}, \mathcal{A}} = \bar{M} \sum_{m \in \{R, O\}} \int_{\mathcal{E}_m(z)} \left[ \pi_i^E \Psi_i^m + (n_i^m - \bar{n}_i^E) \frac{\pi_i^E (1 - \pi_i^E)}{\kappa_E} \Theta_i^m \right] \mathcal{P}_i^m(z) dG_Y(i). \quad (30)$$

The first term is the intensive margin: already-entered households change completed fertility by  $\Psi_i^m$  for each unit of cap relief. The second term is the entry margin: cap relief raises the value of entering, and marginal entrants bring their inside fertility net of the fertility they would have had outside the modeled market. Both terms are gated by the same exposure set. A policy aimed at a slack constraint has zero pass-through in this channel.

At one benchmark point the fertility response and the access gap are the same object.

**Corollary 1** (Equal scaling: one statistic for both margins). *Suppose  $\chi = (1 + k)\kappa q^m/\alpha$ , equal scaling at lifetime prices in mode  $m$ , and the effective cap binds strictly. Then*

$$\Psi_i^m = \frac{\kappa}{\alpha} \frac{\zeta_i^m}{x_i^m} \omega_i^m, \quad \omega_i^m = \frac{\frac{\alpha}{s_i^m} + \frac{q^m}{x_i^m}}{\frac{\vartheta}{(n_i^m)^2} + \frac{\chi^2}{(1+k)(x_i^m)^2} + \frac{\alpha\kappa^2}{(s_i^m)^2}} > 0. \quad (31)$$

Under equal scaling, the normalized access gap  $\zeta_i^m/x_i^m$  is simultaneously the misallocation wedge of the efficiency proposition, the fertility exposure  $\Psi_i^m$  up to the positive weight  $\omega_i^m$ , and the entry response  $\Theta_i^m$ . This is the object the quantitative model measures: Part II reports the distribution of  $\zeta$  among fertile households and the share of the fertility-response mass that sits on  $\zeta > 0$  types.

One pass-through has a closed form that Part II uses. For a collateral-capped owner,  $H_i^O = a_i\rho/[(1-\phi)q]$  with  $\rho = r + \delta + \tau^p$ . Holding the user cost and the old-owner gains wedge fixed, a property-tax increase raises  $\rho$ , lowers the asset price  $P = q/\rho$ , and thereby lowers the down payment per unit of house:

$$\left. \frac{\partial H_i^O}{\partial \tau^p} \right|_{q, \bar{\ell}} = \frac{a_i}{(1-\phi)q} > 0. \quad (32)$$

A property tax, capitalized into the asset price, relaxes the collateral cap by  $a_i/[(1-\phi)q]$  units per point of tax, and the birth response follows from Proposition 2 integrated over the collateral-exposed set. Whether the relief creates births in practice depends on whether exposed households are near their fertility margin when the relief arrives. Part II shows that timing, not exposure, is the binding failure.

The formula is fixed-tenure accounting. If tenure choice is deterministic and the type distribution has density near the rent-own boundary, a policy can also move households across that boundary. Let

$$\Delta_i(z) = W_i^O(z) - W_i^R(z), \quad \mathcal{I}(z) = \{i : \Delta_i(z) = 0\},$$

and suppose  $\nabla_i \Delta_i(z) \neq 0$  on  $\mathcal{I}(z)$ , with unique mode optima and no mass points at kinks. The tenure-switching component is

$$\left. \frac{dN^{\text{switch}}}{dz} \right|_{q, \mathcal{A}} = \bar{M} \int_{\mathcal{I}(z)} \pi_i^E (n_i^O - n_i^R) \frac{\partial_z \Delta_i(z)}{\|\nabla_i \Delta_i(z)\|} f(i) dS(i), \quad (33)$$

where  $f(i)$  is the type density and  $dS(i)$  is unsigned surface measure on the indifference set. The normal  $\nabla_i \Delta_i / \|\nabla_i \Delta_i\|$  is oriented toward the owner region  $\{i : \Delta_i > 0\}$ . The total fixed-price effect is

$$\left. \frac{dN}{dz} \right|_q = \left. \frac{dN}{dz} \right|_{q, \text{tenure}, \mathcal{A}} + \left. \frac{dN^{\text{switch}}}{dz} \right|_{q, \mathcal{A}}. \quad (34)$$

There is no separate entry-boundary term under logit entry because the inside value is continuous at the tenure boundary. The discontinuity is in conditional fertility,  $\pi_i^E(n_i^O - n_i^R)$ . If  $q^O = q^R$ , the

two tenure modes differ only through caps and the boundary fertility jump disappears when the boundary allocation is the same in both modes. With a tax wedge, ownership is the more expensive way to access family space; the switching term is first order, but its sign depends on the fertility jump of boundary households.

The direction of movement across the tenure boundary does not by itself sign the birth response. The integrand also contains  $n_i^O - n_i^R$ . When ownership has the higher effective price, the cheaper tenure is constrained at the boundary, the price gap dominates any continuation-value difference, and the equal-scaling condition holds, the family-size jump is positive only if the within-cap space force also dominates the continuation-value level shift. This is the integral condition derived in the full theory note. Under those sufficient conditions an owner-access policy has a nonnegative switching component, while relaxing the rental cap has a nonpositive one. Outside those conditions, either sign is possible.

## Part II

# Quantitative Lifecycle Model

## 8 Purpose and Timing

The quantitative model embeds the same mechanism in a finite-life economy. It has the standard lifecycle structure: households age, face income risk, save, choose whether to rent or own, choose housing size, decide fertility, and eventually leave bequests. The housing market has one aggregate user cost and one asset price. Tenure matters because it changes the size menu, financing constraints, transaction costs, and tax exposure.

Time is discrete. Households enter adult life at age  $a = 1$ , retire after age  $J_R$ , and die at the end of age  $J$ . The individual state at the beginning of a period is

$$x = (b, z, d, a, n, s). \quad (35)$$

Here  $b$  is liquid wealth,  $z \in \mathcal{Z}$  is idiosyncratic productivity,  $d$  is tenure and housing position,  $a$  is age,  $n$  is an encoded completed-family-size state, and  $s$  is the child-at-home state. Tenure belongs to

$$\mathcal{D} = \{R, O_1, \dots, O_K\}. \quad (36)$$

$R$  denotes renting.  $O_k$  denotes ownership of a house with physical size  $H_k$ , where  $H_1 < \dots < H_K$ . Renters choose housing continuously up to  $\bar{h}^R$ ; owners choose from the discrete ladder. The ladder is the quantitative version of the family-housing segmentation in Part I.

Income for working households is

$$y(a, z) = (1 - \tau_{\text{pay}})w_e a z, \quad z' \sim \Pi^z(\cdot|z), \quad a \leq J_R, \quad (37)$$

and retired households receive pension income  $p^{\text{ret}}$ . The wage is common apart from age and productivity because the quantitative mechanism is housing access, tenure, and lifecycle wealth.

## 9 Preferences and Bequests

Let  $m(n, s)$  be the number of dependent children living at home. Encoded completed family size  $n$  and current dependents differ because children eventually mature. Household needs are

$$\begin{aligned}\bar{c}(n, s) &= \bar{c}_0 + \bar{c}_n m(n, s), \\ \bar{h}(n, s) &= \bar{h}_0 + \bar{h}_{\text{jump}} \mathbf{1}\{m(n, s) \geq 1\} + \bar{h}_n m(n, s).\end{aligned}\tag{38}$$

The first child can create a discrete housing-space jump; additional dependent children raise the required space linearly.

Flow utility is Stone–Geary over consumption and usable housing space, wrapped in CRRA. The owner service premium applies after household needs are removed:

$$u(c, h, d; n, s) = \frac{[(c - \bar{c}(n, s))^{\alpha_c} \mathfrak{h}(h, d; n, s)^{1-\alpha_c}]^{1-\sigma}}{1-\sigma} + \psi_{\text{child}} m(n, s),\tag{39}$$

where

$$\mathfrak{h}(h, d; n, s) = \begin{cases} h^R - \bar{h}(n, s), & d = R, \\ \chi_O [H_k - \bar{h}(n, s)], & d = O_k, \end{cases} \quad \chi_O > 0.\tag{40}$$

Utility is  $-\infty$  when residual consumption or usable space is nonpositive. In market clearing, an owner demands the physical quantity  $H_k$ , not the preference-scaled quantity  $\chi_O [H_k - \bar{h}(n, s)]$ .

At death, households receive warm-glow bequest utility. The house passes to the estate at market value: there is no forced sale, no transaction cost, and no tax at death, so bequeathable wealth is  $W^B = b + PH(d)$ , liquid wealth plus the gross value of any house held. Bequest utility is

$$\mathcal{B}(W^B, n) = \theta_0 (1 + \theta_n n) \frac{(\theta_1 + \max\{W^B, 0\})^{1-\sigma}}{1-\sigma}.\tag{41}$$

This utility is not normalized to zero at a zero estate. Because its scale depends on  $n$ , that level is part of the family-size choice. The parameter  $\theta_n$  lets completed fertility affect the desired estate even after children have left the household.

## 10 Housing, Finance, and Taxes

There is one aggregate housing-services market. The user cost  $q$  and asset price  $P$  satisfy

$$q = (r + \delta + \tau^p)P.\tag{42}$$

Housing supply is upward sloping:

$$H^S(q) = H_0 \left( \frac{q}{\bar{q}} \right)^\eta.\tag{43}$$

The fixed-stock compact model is the limiting case  $\eta = 0$  around the benchmark stock.

Renters choose  $c, h^R, b'$  subject to

$$c + qh^R + b' = Rb + y(a, z), \quad 0 \leq h^R \leq \bar{h}^R, \quad b' \geq 0,\tag{44}$$

where  $R = 1 + r$ . Owners of rung  $H_k$  choose  $c, b'$  subject to the owner branch budget and the borrowing limit described below. Owners pay maintenance and property taxes through  $(\delta + \tau^p)PH_k$ .

At origination, a buyer can finance at most a share  $\phi$  of house value, so the required down payment is

$$(1 - \phi)PH_k. \quad (45)$$

Equivalently, adjusted liquid wealth immediately after purchase must satisfy  $\tilde{b} \geq -\phi PH_k$ . End-of-period saving for every owner, whether a buyer or an incumbent, faces the same debt floor:

$$b' \geq -\phi PH_k. \quad (46)$$

Housing adjustment is costly. A sale of house  $H_k$  incurs the proportional transaction cost  $\psi^s$ , so net sale proceeds are

$$\Omega(H_k) = (1 - \psi^s)PH_k. \quad (47)$$

Transaction costs are the only disposal friction in the computed benchmark: an owner who sells pays  $\psi^s$  per unit of value and faces no tax on accrued gains.

The capital-gains treatment of Part I is specified as an extension and is not part of the computed benchmark. Pricing realized gains requires adding a tax basis  $B^p$  to the owner's state; a sale would then trigger  $\mathcal{T}^g(PH_k, B^p) = \tau^g \max\{PH_k - B^p - \bar{g}, 0\}$  with statutory exclusion  $\bar{g}$ , net proceeds would fall to  $\Omega(H_k) - \mathcal{T}^g(PH_k, B^p)$ , and stepped-up basis at death would forgive the accrued gain. The benchmark state vector (35) deliberately excludes  $B^p$ : the old-owner lock-in wedge  $\bar{\ell}$  is priced analytically in Part I, and the counterfactual section instead bounds its quantitative content with a direct retention-margin experiment.

## 11 Fertility and Child Aging

Fertility is chosen during the fertile window

$$a \in \{A_f^{\text{start}}, \dots, A_f^{\text{end}}\}.$$

A childless household chooses

$$n' \in \mathcal{N} = \{0, 1, \dots, \bar{n}\}. \quad (48)$$

The option  $n' = 0$  means no birth in the current period. A positive choice fixes completed fertility forever and moves the household into the first dependent-child state. A household that reaches the end of the fertile window without a positive choice remains childless.

In the computation,  $\mathcal{N} = \{0, 1, 2\}$  and the last state is the open-ended highest family-size category. The completed-fertility moment matched to the CPS is reported as  $2\mathbb{E}[n]$ . This household-to-woman conversion is applied only to that reported moment. Child needs and bequest utility use the encoded state  $n$ , while the housing-size target maps  $n = 1$  to the 1–2 group and  $n = 2$  to the 3+ group. The distinction is a unit convention and does not create an additional fertility choice.

Let  $s = 0$  denote no dependent children and let  $s = 1, \dots, S_c$  denote dependent-child stages. After a positive fertility choice, the household enters  $s = 1$ . Children age stochastically through the stages according to  $\Pi^c$ . While children are at home,  $m(n, s) = n$ ; after maturity,  $m(n, s) = 0$ . Completed fertility still matters through bequest utility.

## 12 Household Problem

Within the period, the household first faces fertility if eligible, then chooses tenure and housing position, then chooses consumption, rental housing, and saving. Let the continuation value after income risk and child aging be

$$\bar{V}_{a+1}(b', d', z, n, s) = \sum_{z'} \sum_{s'} \Pi^z(z'|z) \Pi^c(s'|s, n) V_{a+1}(b', z', d', n, s'). \quad (49)$$

The terminal value is

$$V_{J+1}(b, z, d, n, s) = \mathcal{B}(W^B, n).$$

Conditional on renting, the branch value is

$$\begin{aligned} V_a^R(b, z, n, s) = \max_{c, h^R, b'} & u(c, h^R, R; n, s) + \beta \bar{V}_{a+1}(b', R, z, n, s) \\ \text{s.t.} & c + qh^R + b' = Rb + y(a, z), \\ & 0 \leq h^R \leq \bar{h}^R, \quad b' \geq 0. \end{aligned} \quad (50)$$

Conditional on owning rung  $H_k$  after any adjustment, the branch value is

$$\begin{aligned} V_a^{O_k}(\tilde{b}, z, n, s) = \max_{c, b'} & u(c, H_k, O_k; n, s) + \beta \bar{V}_{a+1}(b', O_k, z, n, s) \\ \text{s.t.} & c + (\delta + \tau^p)PH_k + b' = R\tilde{b} + y(a, z), \\ & b' \geq -\phi PH_k. \end{aligned} \quad (51)$$

The debt floor is the same for a buyer and an incumbent. Purchase feasibility is screened separately through the down-payment condition before this branch is evaluated.

Let  $H(R) = 0$  and  $H(O_k) = H_k$ . The liquid wealth available after housing adjustment is

$$\tilde{b}(b, d, d') = b + \mathbf{1}\{d \in \mathcal{O}, d' \neq d\} \Omega(H(d)) - \mathbf{1}\{d' \in \mathcal{O}, d' \neq d\} PH(d'). \quad (52)$$

A seller collects net proceeds, a buyer pays the full purchase price, and a household that keeps its position pays nothing.

The tenure branch value for feasible  $d'$  is

$$\tilde{V}_a^T(d'|b, z, d, n, s) = \begin{cases} V_a^R(\tilde{b}(b, d, R), z, n, s), & d' = R, \\ V_a^{O_k}(\tilde{b}(b, d, O_k), z, n, s), & d' = O_k. \end{cases} \quad (53)$$

Tenure and owner-rung choices have Type-I extreme-value shocks with scale  $\kappa_T$ :

$$\pi_a^T(d'|x) = \frac{\exp\{\tilde{V}_a^T(d'|x)/\kappa_T\}}{\sum_{\hat{d} \in \mathcal{D}(x)} \exp\{\tilde{V}_a^T(\hat{d}|x)/\kappa_T\}}, \quad (54)$$

and the inclusive tenure value is

$$V_a^T(x) = \kappa_T \log \sum_{\hat{d} \in \mathcal{D}(x)} \exp\{\tilde{V}_a^T(\hat{d}|x)/\kappa_T\}. \quad (55)$$

At the estimated value  $\kappa_T = 0$ , these expressions mean their deterministic limit: the branch with the highest value is chosen with probability one, and a tie is resolved by the fixed ordering of feasible branches.

For a childless household in the fertile window, the value of fertility option  $n'$  is

$$\tilde{V}_a^F(n'|b, z, d) = V_a^T(b, z, d, n', s(n')), \quad s(n') = \mathbf{1}\{n' > 0\}. \quad (56)$$

With Type-I extreme-value shocks of scale  $\kappa_F$ ,

$$\pi_a^F(n'|b, z, d) = \frac{\exp\{\tilde{V}_a^F(n'|b, z, d)/\kappa_F\}}{\sum_{\hat{n} \in \mathcal{N}} \exp\{\tilde{V}_a^F(\hat{n}|b, z, d)/\kappa_F\}}. \quad (57)$$

The value before the fertility draw is the corresponding inclusive value:

$$V_a(b, z, d, 0, 0) = \kappa_F \log \sum_{\hat{n} \in \mathcal{N}} \exp\{\tilde{V}_a^F(\hat{n}|b, z, d)/\kappa_F\}. \quad (58)$$

Outside the fertile window, or after a positive fertility choice, this stage is degenerate and  $V_a = V_a^T$ .

### 13 Stationary Equilibrium

Let  $g^-$  be the stationary distribution at the beginning of a period, before the current fertility, tenure, and housing choices. Applying those choices to  $g^-$  produces the post-decision distribution  $g^+$ . Flow utility, current housing demand, and age-specific moments are measured with  $g^+$ ; the law of motion maps  $g^+$  into next period's  $g^-$ . Adult mass is normalized to one:

$$\int dg^-(x) = \int dg^+(x) = 1. \quad (59)$$

This normalization fixes the demographic envelope and lets fertility change household composition, tenure, and housing demand. It is not, by itself, a demographic closure for total births: the entry inflow of new adults is a fixed constant and births do not feed it. Quantitative counterfactuals therefore report per-capita objects — completed fertility, childlessness, tenure — at fixed population. The entry and scale margin of Part I is reinstated on top of the solved model by an outer fixed point described in Section 18.1, and its results are reported separately from the fixed-population results.

**Definition 1** (Stationary equilibrium). *A stationary equilibrium is a user cost and asset price  $(q, P)$ , value functions and policies, tenure and fertility probabilities, stationary distributions  $(g^-, g^+)$ , and aggregate housing quantities  $(H^D, H^S)$  such that:*

1. *households solve the branch problems (50)–(51), tenure choice is given by (54), fertility choice is given by (57), and terminal utility is (41);*
2. *current fertility and tenure policies map  $g^-$  into  $g^+$ , and the income transition, child aging, deterministic aging, death, and entry of new adult households map  $g^+$  back into  $g^-$ ;*
3. *aggregate physical housing demand is*

$$H^D = \int \left[ \mathbf{1}\{d = R\}h^R(x) + \sum_{k=1}^K \mathbf{1}\{d = O_k\}H_k \right] dg^+(x); \quad (60)$$

4. *the housing market clears and the user-cost relation holds:*

$$H^D = H^S(q), \quad q = (r + \delta + \tau^p)P; \quad (61)$$

5. *payroll taxes, property taxes, capital-gains taxes, pensions, and any rebates satisfy the government accounting rule specified for the calibration or counterfactual.*

## 14 Computation

One model period is four years. Period  $j = 0, \dots, 16$  begins at age  $18 + 4j$ ; the last interval covers ages 82–85. Retirement begins at 66, and the last period in which a birth can occur begins at 42. Annual objects are converted before solution:

$$R = (1 + r_{\text{ann}})^4, \quad \beta = \beta_{\text{ann}}^4, \quad \delta = 1 - (1 - \delta_{\text{ann}})^4, \quad \tau^p = 4\tau_{\text{ann}}^p.$$

Income and consumption-flow quantities are expressed in four-year units; housing remains in rooms.

The wealth grid has  $N_b = 120$  nodes on  $[-35, 100]$  with power spacing near zero. Income takes values  $z \in \{0.6, 0.8, 1.0, 1.2, 1.4\}$  with invariant weights  $\pi_z = (0.1, 0.2, 0.4, 0.2, 0.1)$  and transition matrix  $\Pi^z = 0.85I + 0.15\mathbf{1}\pi'_z$ . One dependent-child stage has an expected duration of 18 years, giving a four-year maturity probability of  $2/9$ . Entrants have mass  $1/J$  each period. The balanced pension equals payroll-tax revenue per retiree in the stationary age distribution.

Value functions are computed by backward induction. Golden-section search solves saving in each branch, and Howard iteration accelerates evaluation. The stationary distribution follows by forward iteration, while damped fixed-point iteration and a scalar bracketing refinement solve the housing market. The intended market-residual tolerance is  $10^{-4}$ ; strict rejection of candidates above that tolerance is part of the corrected production run.

Because tenure and fertility choices are close to deterministic at the calibrated point, moments built on buy thresholds and the borrowing constraint move with the wealth grid, and grid resolution matters more here than in a standard lifecycle model. An earlier calibration at  $N_b = 60$  re-evaluated 18 percent worse at  $N_b = 120$  at the same parameters; part of its fit was grid noise. The July snapshot was searched at  $N_b = 120$  and re-evaluated at  $N_b = 240$ . The corrected estimation will use the same search and verification grids.

## 15 Calibration

### 15.1 Externally set objects

Prices, finance, demography, and the housing menu are set outside the moment-matching loop.



Table 1: Externally set parameters

| Object                          | Value                                             | Source or rule                                                                                                                    |
|---------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Interest rate $r$               | 4% per year                                       | standard                                                                                                                          |
| Depreciation $\delta$           | 2% per year                                       | standard                                                                                                                          |
| Property tax $\tau^p$           | 1% per year                                       | U.S. average effective rate                                                                                                       |
| Financed share $\phi$           | 0.80                                              | 20% down payment                                                                                                                  |
| Sale cost $\psi^s$              | 0.06                                              | transaction-cost evidence                                                                                                         |
| Risk aversion $\sigma$          | 2                                                 | standard                                                                                                                          |
| Bequest shift $\theta_1$        | 0.01                                              | near-linear warm glow                                                                                                             |
| Payroll tax $\tau_{\text{pay}}$ | 0.179                                             | pension balances in the stationary distribution                                                                                   |
| Income profile $e_a$            | 0.65 at 18, rises to 1.0 at 34, declines after 58 | intended profile; age-18 cell under correction                                                                                    |
| Owner menu $\{H_k\}$            | $\{2, 4, 6, 8, 10\}$ rooms                        | room-count support, ACS                                                                                                           |
| Renter cap $\bar{h}^R$          | 6.0 rooms                                         | 90th-percentile rental size rule of <a href="#">Greaney et al. (2025)</a> , measured in the matched ACS sample; 6.5 as robustness |
| Supply $(H_0, \eta)$            | $(4, 1)$                                          | unit-elastic benchmark                                                                                                            |
| Entry wealth                    | PSID wealth-to-income ratios at age 22–25         | first stage                                                                                                                       |
| Discount floor                  | $\beta_{\text{annual}} \geq 0.94$                 | declared restriction                                                                                                              |

The renter cap is the quantitative version of the compact model’s  $\bar{h}^R$ . I measure it from the size distribution of rental units rather than estimate it. The calibrated fit is nearly flat within half a room of the measured value and deteriorates beyond, so the data cannot separate 6.0 from 6.5 but do reject anything further out. The discount floor binds in the estimation below, so  $\beta$  is in effect a restriction rather than an estimate.

## 15.2 Estimated parameters

Thirteen parameters are estimated by simulated method of moments: the discount factor  $\beta$ , the consumption share  $\alpha_c$ , the adult floors  $\bar{c}_0$  and  $\bar{h}_0$ , the child cost parameters  $\bar{c}_n$ ,  $\bar{h}_{\text{jump}}$ ,  $\bar{h}_n$ , and  $\psi_{\text{child}}$ , the choice scales  $\kappa_F$  and  $\kappa_T$ , the owner service multiplier  $\chi_O$ , and the bequest parameters  $\theta_0$  and  $\theta_n$ . The objective is a weighted sum of squared moment gaps, minimized by a global differential-evolution stage followed by self-reseeding local polish chains on a cluster, at  $N_b = 120$ .

Table 2: July 9 estimates, search bounds, and bound status (provisional)

| Parameter               | Estimate | Bounds         | Status                          |
|-------------------------|----------|----------------|---------------------------------|
| $\beta_{\text{annual}}$ | 0.940    | [0.940, 0.995] | at lower (declared restriction) |
| $\alpha_c$              | 0.620    | [0.40, 0.95]   | interior                        |
| $\bar{c}_0$             | 1.279    | [0.08, 1.28]   | at upper                        |
| $\bar{c}_n$             | 0.116    | [0.05, 1.50]   | interior                        |
| $\bar{h}_0$             | 1.000    | [1.0, 6.0]     | at lower                        |
| $\bar{h}_{\text{jump}}$ | 2.120    | [0.05, 2.50]   | interior, high                  |
| $\bar{h}_n$             | 1.382    | [0.02, 2.00]   | interior                        |
| $\psi_{\text{child}}$   | 0.349    | [0, 0.35]      | at upper                        |
| $\kappa_F$              | 1.019    | [1.0, 12.0]    | at lower                        |
| $\kappa_T$              | 0.000    | [0, 0.12]      | at lower                        |
| $\chi_O$                | 1.150    | [0.40, 1.15]   | at upper                        |
| $\theta_0$              | 0.0015   | [0, 2.0]       | at lower                        |
| $\theta_n$              | 0.972    | [0, 1.50]      | interior                        |

Eight of thirteen parameters sit at or against the search box recorded with the July run. This is a feature of the provisional objective, not a final identification result. The same production bounds must now be placed in one named source profile before the corrected search. The near-zero bequest weight also leaves  $\theta_n$  weakly identified in local sensitivity calculations.

The two choice scales land at corners with different meanings. The tenure scale  $\kappa_T$  estimates to zero: the data prefer deterministic tenure choice, and the smoothing device does no work. The fertility scale  $\kappa_F$  estimates to its floor, where the birth choice is also near-deterministic, but there the choice structure carries the entire fertility block. At a near-degenerate logit, low-income types have birth probabilities of exactly zero, and fertility moments inherit sensitivity to the discreteness of the wealth grid. The grid protocol of Section 14 guards against the discreteness; whether taste smoothing belongs in these choices at all is a question the estimation cannot settle.

### 15.3 Target moments and fit

The calibration targets fourteen moments: completed fertility and childlessness from the CPS (the June 2024 supplement cells for women 40–44), tenure and housing-size moments by age and family status from a matched ACS metropolitan sample, and wealth moments from the PSID. The table reports every target with its model moment, gap, weight, and loss contribution. The total loss at the reported point is 6.31.

Table 3: July 9 target moments and model fit ( $N_b = 120$ ; provisional loss 6.31)

| Moment                                                | Target | Model | Gap    | Weight | Contribution |
|-------------------------------------------------------|--------|-------|--------|--------|--------------|
| Old-age ownership rate (65–75)                        | 0.764  | 0.893 | +0.128 | 160    | 2.64         |
| Ownership rate 25–34                                  | 0.341  | 0.226 | −0.115 | 80     | 1.05         |
| Owner–renter mean rooms (no children at home, 30–55)  | 2.419  | 2.151 | −0.268 | 12     | 0.86         |
| Family ownership gap                                  | 0.168  | 0.298 | +0.130 | 45     | 0.77         |
| Young renter liquid wealth / income (25–35)           | 0.179  | 0.353 | +0.173 | 12     | 0.36         |
| Aggregate ownership rate                              | 0.575  | 0.531 | −0.045 | 100    | 0.20         |
| Renter mean rooms (no children at home, 30–55)        | 3.805  | 3.655 | −0.150 | 6      | 0.14         |
| Childlessness rate                                    | 0.188  | 0.260 | +0.072 | 20     | 0.10         |
| Rooms gap, 3+ vs. 1–2 children (30–55)                | 0.368  | 0.269 | −0.099 | 8      | 0.08         |
| Old parent–childless nonhousing wealth gap            | 1.007  | 0.842 | −0.166 | 2      | 0.05         |
| Housing increment at first birth (rooms)              | 0.664  | 0.618 | −0.046 | 14     | 0.03         |
| Completed fertility                                   | 1.918  | 1.949 | +0.031 | 20     | 0.02         |
| Old nonhousing wealth / income, median (65–75)        | 2.230  | 2.328 | +0.097 | 0.8    | 0.01         |
| Owner share with rooms $\geq 6$ (no children at home) | 0.596  | 0.593 | −0.003 | 25     | 0.00         |

Gaps are rounded for display. Contributions and the reported loss use the unrounded model moments. Several tenure and housing rows will change when current choices are measured with the post-decision distribution; the first-birth housing row also omits the same-period purchase margin in this snapshot.

The July snapshot is close to the fertility level and several size-distribution moments. Old-age ownership is too high: the model keeps 89 percent of 65–75 households in owned housing against 76 percent in the data, because nothing in the model pushes an old owner out of her house. There are no health shocks and no mortality-linked downsizing, so retention is close to absorbing. Ownership at 25–34 is too low (0.23 against 0.34), while young renter wealth is too high (0.35 against 0.18). Both moments are directly affected by the pending timing and income-profile corrections, so the current gaps do not identify a structural failure on their own. Childlessness also overshoots (0.26 against 0.19). That miss remains a substantive feature of the current choice structure, but its quantitative size must be reassessed at the corrected baseline.

Re-evaluated at  $N_b = 240$  at the same parameters, the loss rises from 6.31 to 6.84, and the drift concentrates in the family ownership gap and the old-age ownership rate, both of which move further from their targets at the finer grid. This comparison records the numerical drift in the July snapshot. The corrected calibration will report the same 120/240 comparison.

## 16 The July 9 Equilibrium Snapshot

The lifecycle pattern is descriptive of the provisional snapshot. In particular, the plotted ownership stock uses the inherited tenure state and therefore records current purchases one period late. The corrected run will replace this figure rather than reinterpret it.

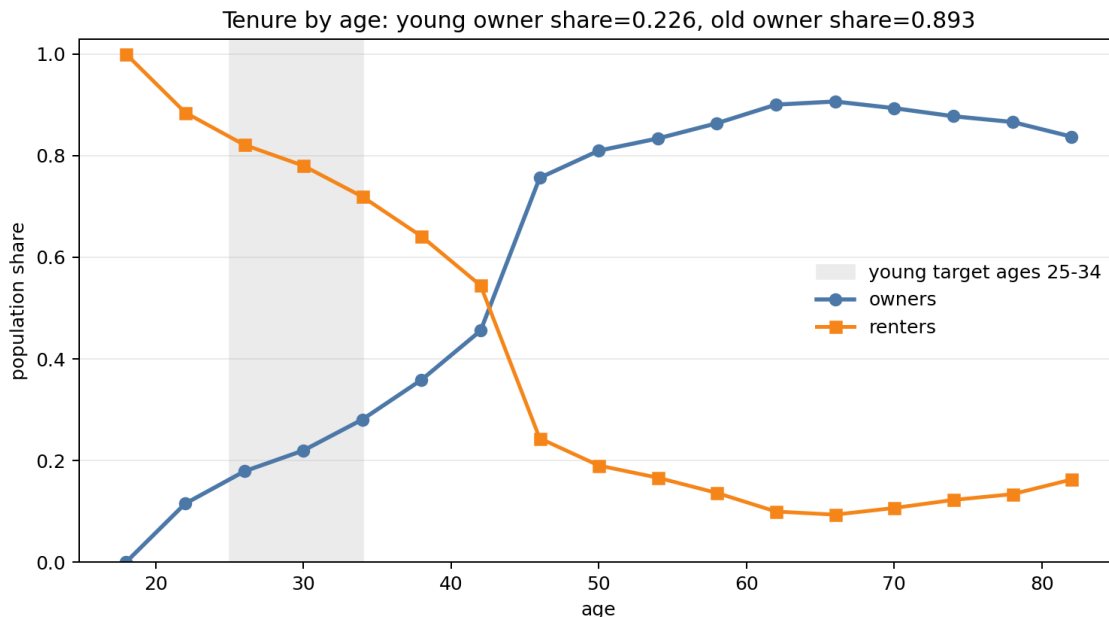


Figure 2: Ownership over the lifecycle in the July 9 snapshot. The steep rise near the end of the fertile window and high old-age ownership are visible, but current purchases are recorded one model period late in this version.

Renting households with children sit against the rental cap, while owners use the discrete room ladder. Family-sized space is scarce through availability: within a size cell, the model rent carries no separate premium. The snapshot also places many young renters near zero liquid wealth. These features remain useful diagnostics, but their age profiles and aggregate weights must be regenerated after the timing correction.

## 17 The Mechanism Quantified

Part I's argument runs through one object, the access gap  $\zeta$ : the amount by which a constrained household's marginal valuation of family space exceeds its market user cost. Under equal scaling, the corollary to Proposition 2 says the same gap prices misallocation, fertility exposure, and entry. The July snapshot gives a first measurement of that gap.

The measured gap is large. Among fertile-age renting households pinned at the rental cap, the mean access gap is 0.248 goods units, 1.28 times the market user cost, so capped renters value a marginal room at 2.3 times what the market charges for it. Weighting fertility responses by exposure as in Proposition 2, 73 percent of the model's fertility-response mass sits on households with  $\zeta > 0$ . Within the July diagnostics, the share is similar across grid resolutions and owner menus. Its level remains provisional because the exposure weights will change when current housing choices are measured contemporaneously.

The model places many purchases near the fertility deadline. The purchase region expands in the last fertile period, when the option value of waiting to have children expires. Buyers in this region hold wealth above the required down payment. In the current model, housing access therefore does not operate only by forcing households to save toward a distant threshold. It also compresses births

and purchases into a joint decision near the deadline. The corrected run will determine how much of this pattern survives contemporaneous measurement.

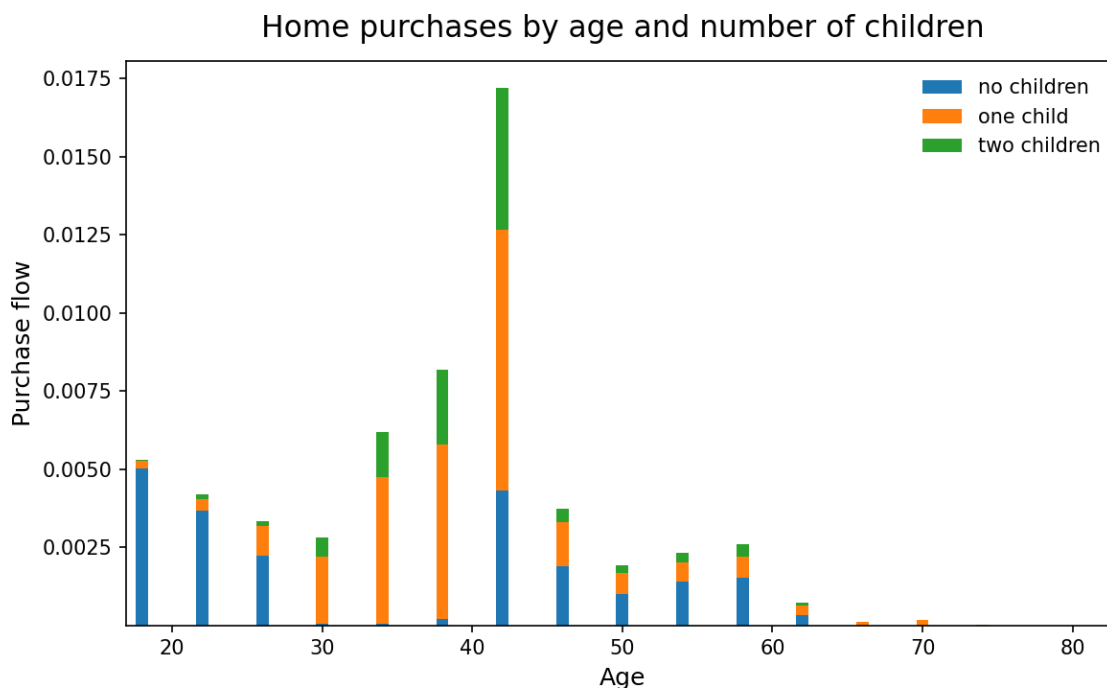


Figure 3: Purchase flows by age and family size in the July 9 snapshot. Buying is concentrated near the end of the fertile window and is birth-linked.

The compact model’s misallocation proposition is an accounting statement about who holds family-capable space, and it can be checked against the data. The table reports the allocation of the owner stock with six or more rooms, in the calibrated equilibrium and in the matched ACS sample.

Table 4: Who holds the family-sized owner stock (rooms  $\geq 6$ )

| Holder                                      | Data (ACS) | Model |
|---------------------------------------------|------------|-------|
| Age 18–29                                   | 3.4%       | 0.4%  |
| Age 30–45                                   | 24.8%      | 12.7% |
| Age 46–64                                   | 43.3%      | 50.5% |
| Age 65+                                     | 28.5%      | 36.5% |
| Held by households 46+                      | 71.8%      | 87.0% |
| Held by households with no children at home | 54.9%      | 72.5% |

The concentration is in the data, not only in the model. In the ACS, 72 percent of the family-sized owner stock is held by post-fertile households and 55 percent by households with no children at home. The model has the same ordering and overstates the concentration among older households. Its model column will be recomputed after the timing correction. The data column already shows the allocation problem that motivates the model.

A preliminary retention experiment moves prices by less than one percent and completed fertility

by about one hundredth. The result mainly reveals that the current old households are close to a retain-everything corner: without a health, mortality, or downsizing margin, few of them respond to a disposal wedge. I treat this exercise as a runtime pattern, not as an estimate of the old side of the allocation wedge.

## 18 Preliminary Policy Patterns

The instruments follow the access margins of the compact model: a holding tax on large owner units (the capitalization channel of eq. 32), birth-linked down-payment grants restricted to family-sized rungs (the financing gap  $\zeta^{O,F}$ ), variation in the rental cap  $\bar{h}^R$ , a retention wedge on old owners, and a revenue-linked package combining the tax and the grant. These exercises describe the July diagnostic runtime, not final policy estimates. The grant enters both household values and the forward distribution in that runtime. Prices re-clear in every fixed-population experiment, but all cases will be rerun after re-estimation.

The current tax experiment produces capitalization and a small birth response. A two percent holding tax on large units lowers their asset price by 12 percent. Capitalization is similar across the grids, moment weights, and owner menus that have been checked, while completed fertility rises by +0.004 to +0.009. As eq. 32 predicts, down payments fall. In this snapshot, exposed households buy near the fertility deadline with wealth already above the threshold, so cheaper access changes which house they buy more than whether they have children.

The grant runtime produces a dose-response. A down-payment grant paid at a birth, restricted to family-sized rungs, raises completed fertility by +0.01, +0.13, +0.26, and +0.49 at grant sizes 0.1, 0.4, 0.58, and 1.0, in units of mean annual income. In the current solution the access instrument has a larger response because it is conditional on the birth itself: it closes the financing gap at the moment the household is on its fertility margin. The conditioning also distorts. Small grants delay pre-birth purchases, since households near a birth wait for the subsidy, and young ownership falls from 0.23 to 0.09 at the smallest grant. The distortion fades at larger grant sizes, but any conditional instrument inherits the margin.

The combined experiment follows both sides of Part I's reallocation. The two percent tax on large homes funds the birth-linked grant; the tax lowers incumbents' retention value and the price of large units by 11 percent, and the grant closes the young side's financing gap at the birth margin. At the July calibration the package raises completed fertility by +0.15 and is revenue-funded. This magnitude is provisional.

### 18.1 The entry and scale margin

The results above hold population fixed. The entry margin of Part I is reinstated by an outer fixed point around the solved model: the policy changes household values, values move the entry probability against the outside option and the number of city-born future entrants, the implied stationary population scale  $S$  multiplies aggregate housing demand, and the price re-clears against the supply curve. Households never see  $S$  directly; population size operates through the price of space. The outside value and entry scale are recovered at the benchmark so that the baseline is exactly  $S = 1$ . The recovered conventions are documented rather than estimated, and the binding one is the assumption that city-born children become future entrants one-for-one.

These results use  $N_b = 120$  at the July calibration. The fixed-population column re-solves each case

inside the same protocol. The grant and package rows carry a few-percent price imprecision from the outer loop’s fast solves.

Table 5: July 9 policy runtime with the entry margin on ( $N_b = 120$ ; provisional)

| Policy                    | $\Delta\text{TFR}$ , fixed pop. | $\Delta\text{TFR}$ , intensive | Scale $S$ | Total births    |
|---------------------------|---------------------------------|--------------------------------|-----------|-----------------|
| Holding tax 2%            | +0.004                          | +0.002                         | 1.003     | +0.4%           |
| Grant 0.4, rungs $\geq 6$ | +0.119                          | +0.083                         | 1.038     | +8.2%           |
| Package (tax + grant)     | +0.162                          | +0.086                         | 1.040     | +8.6%           |
| Rental cap 6.5            | −0.006                          | −0.009                         | 1.004     | $\approx 0.0\%$ |
| Retention wedge 0.2       | +0.006                          | +0.000                         | 1.007     | +0.7%           |

In the current runtime, the grant raises population scale and the resulting price response offsets part of its fixed-population fertility effect. Total births nevertheless rise in the grant and package rows. The holding tax and retention wedge remain close to neutral. These patterns distinguish per-family fertility from births generated in the modeled market. Whether a local birth gain represents creation or relocation remains outside the one-market model.

## 19 Current Repairs and Re-estimation

The first repair concerns timing. The Bellman problem gives a household its new tenure and house in the current period, but the July distribution code uses inherited tenure for market clearing and several moments. The corrected operator will form a post-decision distribution for current demand and moments, while preserving the next-period distribution for state transitions. The same change will add current purchases to the housing response measured at a first birth.

The second repair covers the first earnings cell. The intended profile begins at 0.65 of peak earnings at age 18, but the uncovered first cell in the July runtime remained at one. The new profile will cover every age explicitly. Neither correction has yet been re-estimated, and I do not infer the new fit from their mechanical directions.

The production run also needs one recorded configuration for the five owner rungs, parameter bounds, target definitions, and the 120/240 wealth grids. A candidate will be ranked only if it meets the strict market-clearing tolerance. The forward-map grant correction used for the July policy runtime will be carried into the main package, and the entry and retention drivers will be included in the same reproducible path. The quantitative sequence is therefore fixed: verify the two timing identities, re-estimate at  $N_b = 120$ , verify at  $N_b = 240$ , regenerate the diagnostic packet, and rerun the policy table.

The paper-side specification is already aligned with the estimated object in three places. Owner utility applies  $\chi_O$  to usable space  $H_k - \bar{h}(n, s)$ , bequest utility retains its family-size-dependent level, and all owners face the same debt floor. Those changes describe the existing estimated preferences and finance rule; they are not new mechanisms.

## 20 Limitations

The model remains deliberately incomplete after these repairs. Old-age ownership is too high because health, mortality, and other reasons for downsizing are absent. Fertility has no age-dependent fecundity, so waiting is costless until the final fertile period. Both omissions matter for the timing and stock-allocation questions at the center of the paper.

The computed benchmark prices old-owner retention through transaction costs only. The capital-gains and basis mechanism of Part I is not in the quantitative state space. The supply curve is unit-elastic, while the compact misallocation result holds the stock fixed. Finally, the entry exercise relies on an externally recovered outside value and on the convention that local births become future entrants. The entry table is therefore a decomposition, not an estimated migration response.

## 21 Conclusion

Children use housing space, and access to that space differs sharply by tenure. A rental-size limit raises the shadow price of room for a constrained household. Ownership relaxes the size limit but introduces a down payment. Retention frictions can keep large homes with older households whose current use of space is low. The compact model places these margins in one goods-equivalent access gap and shows how the same gap enters the private price of children.

The July quantitative snapshot suggests that this mechanism is active. Many households near the fertility margin value additional space above its user cost, purchases bunch late in the fertile window, and the data show that older households hold most family-sized owner housing. The size of these effects is not yet settled. Current tenure timing and the first earnings cell affect targeted moments and require a new calibration.

The immediate goal is therefore narrower than a new model extension: align the distribution with the household timing, restore the intended income profile, freeze one production configuration, and re-estimate. The theory and the data fact organize that exercise. The corrected fit and policy table will determine how much quantitative weight the mechanism can carry.



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